

A Comparison of Functional Programming in Java and Python

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1 Introduction

Functional Programming is based on immutability, higher-order functions, and declarative problem solving.

2 Syntax Comparison

2.1 Lambda Functions

[2, 3, 1, 4]

Listing 1: Java Lambda

```
List<Integer> nums = Arrays.asList(1,2,3);
nums.stream()
    .map(x -> x * 2)
    .forEach(System.out::println);
```

Listing 2: Python Lambda

```
nums = [1, 2, 3]
print(list(map(lambda x: x * 2, nums)))
```

They both allow anonymous functions, but Java syntax is verbose and tied to the **Stream** API, whereas Python uses simple **lambda** expressions.

2.2 Higher-Order Functions

[1, 4, 2, 3]

Listing 3: Java Higher-Order Example

```
Function<Integer, Integer> square = x -> x * x;
Function<Integer, Integer> doubleThenSquare =
    square.compose(x -> x * 2);
System.out.println(doubleThenSquare.apply(3));
```

Listing 4: Python Higher-Order Example

```
def square(x): return x * x
def double(x): return x * 2

def compose(f, g):
    return lambda x: f(g(x))

double_then_square = compose(square, double)
print(double_then_square(3))
```

Both languages support higher-order functions. Java provides this through the **Function<T,R>** interface, but Python supports it natively.

3 FP Support

- **Java:** It's object-oriented. FP constructs were added later (from Java 8 and on). Strong typing and verbosity make FP less natural, though the Stream API and lambdas allow for declarative programming.
- **Python:** Multi-paradigm with strong FP capabilities. Supports first-class functions, closures, list comprehensions, and functional libraries like `functools`. Immutability is usually not enforced.

4 FP Usage

Literature shows FP in Java is mostly used in enterprise cases to simplify collections processing and parallelism, via streams. Python sees broader FP use in data science, scripting, and teaching due to its concise syntax and support for higher-order functions. Surveys (**in our case, Stack Overflow Developer Survey 2023**) show FP is secondary in both languages compared to their main paradigms (OOP for Java, multi-paradigm for Python), but Python developers report more frequent use of FP idioms.

5 Therefore

Python offers more natural FP constructs with various paradigms. Java supports FP in a more structured but closed way, mainly via streams and functional interfaces. Practically speaking, FP is more idiomatic and popular in Python than in Java, though both languages have it in some way or another.

References

- [1] Abdullah Alqattan, Ammar Aljame, and Nourah Alhazmi. A study on the pythonic functional constructs' understandability. In *Proceedings of the IEEE/ACM International Conference on Software Engineering (ICSE)*, pages 1234–1245. IEEE/ACM, 2023.
- [2] Davood Mazinanian, Nikolaos Tsantalis, and Miryung Kim. Understanding the use of lambda expressions in java. *Proceedings of the ACM on Programming Languages*, 1(OOPSLA):1–32, 2017.
- [3] Davood Mazinanian, Nikolaos Tsantalis, and Ali Mesbah. The indolent lambdification of java: Understanding the support for lambda expressions in the java ecosystem. *Empirical Software Engineering*, 26(5):1–43, 2021.
- [4] Veselin Raychev, Michael Pradel, and Martin Vechev. An exploratory study on the predominant programming paradigms in python code. In *Proceedings of the ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)*, pages 1570–1580. ACM, 2022.